

CLARK ATLANTA UNIVERSITY

*School of Arts and Sciences
Department of Public Administration*

CPAD 504

Economics and Data Science for Public Administrators

Revised Syllabus | Gateways 2026 Course Redesign

Instructor	Oyebade Kunle Oyerinde, PhD
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Credit Hours	3
Semester / Meeting Time	TBA 6:00 p.m.-8:30 p.m.
Delivery / Learning Platforms	In-person with Canvas, Jetstream2, CloudBank, and RStudio

This revision integrates the FacultyHack@Gateways 2026 course redesign: cloud-based R analysis, responsible AI-assisted learning, ORNL datasets, and reproducible policy-data storytelling.

Course Description

This graduate course equips public administrators with economic reasoning and data-science practices for evidence-based decision-making and effective policy implementation. Students integrate microeconomic and macroeconomic concepts with applied data analysis to identify organizational choices, analyze the provision and production of goods, evaluate consumer and producer surplus and deadweight loss, and construct policy options for market failures. Using R and RStudio in Jetstream2 and CloudBank environments, students analyze public data, create reproducible visualizations, and communicate findings through clear data stories. The course also emphasizes ethical, transparent, and responsible use of data and AI-assisted tools in public administration.

Course Objectives

- Understand core economic principles and the ethical use of data science in public administration.
- Explain rationales for government intervention and public provision of goods.
- Apply interdisciplinary economic and analytical skills to formulate and evaluate public policy.
- Use data, visualization, and economic logic to assess policy alternatives and resource-allocation choices.

Student Learning Outcomes

- Identify and explain microeconomic, macroeconomic, and data-science concepts applicable to organizational decision-making (SLO 1; NASPAA Competency 3).
- Analyze economic data using R, Python, and Excel in cloud-supported environments (SLO 2; NASPAA Competency 3; DSLO 1-2).
- Communicate evidence through data storytelling and visualizations produced in R and/or Python (SLO 3; NASPAA Competency 5; DSLO 6).
- Construct and defend alternative policy options to remedy a market failure using problem sets and written analysis (SLO 4; NASPAA Competency 2).
- Assess and communicate ethical considerations in data science and AI-supported analysis (SLO 5; NASPAA Competencies 3 and 5; DSLO 5-6).

Required Readings and Learning Resources

- Callan, Scott J., and Janet M. Thomas. Environmental Economics and Management: Theory, Policy, and Application.
- Robbins, Donijo. Handbook of Public Sector Economics.
- Ostrom, Vincent. The Meaning of American Federalism: Constituting a Self-Governing Society.
- Introduction to Data Science (e-book), selected chapters 1-11.
- Publication Manual of the American Psychological Association, 6th ed. (APA, 2010), and instructor-posted resources on algorithmic bias, data privacy, and responsible AI.

Science Gateway and Data Resources

Students will use the following resources as directed by the instructor. Account setup, access requirements, and approved datasets will be provided through Canvas.

- Jetstream2 and CloudBank Hub for cloud-based R/RStudio analysis and reproducible projects.
- SGX3 Gateways LLMs (llm.jetstream-cloud.org) and other instructor-approved AI tools for idea generation, source discovery, code explanation, and code refinement.
- Datasets from the Oak Ridge National Laboratory data portal: doi.ccs.ornl.gov/datasets.

- Science Gateways resource directory: sciencegateways.org/resources/browse.

Teaching and Learning Methods

- Lecture, class discussion, and interdisciplinary/cooperative learning.
- Problem-based learning and policy case analysis.
- Hands-on R/RStudio labs in Jetstream2 and CloudBank.
- Student presentations, writing assignments, and data-storytelling exercises.
- Canvas for modules, announcements, feedback, and assignment submission.

Responsible AI and Reproducible Analysis Policy

AI tools are permitted only as learning assistants when explicitly allowed for an assignment. Students remain responsible for the accuracy, originality, documentation, and interpretation of all work submitted.

- Disclose approved AI assistance in a brief appendix or code comment, including the tool used and how its output was verified or revised.
- Do not submit AI-generated prose, analysis, data, code, citations, or visualizations as your own without meaningful review, revision, attribution, and instructor authorization.
- Validate every AI-assisted claim, source, formula, and code output. Fabricated sources, unverified calculations, or misrepresented AI use are academic-integrity violations.
- Submit reproducible work: code, data-source information, outputs, and a concise explanation of analytical decisions, as required by the assignment.

Canvas and Assignment Submission

Canvas is the official course site for materials, announcements, assignments, examinations, feedback, grades, and course communications. Submit work only in its designated Canvas assignment folder by the stated deadline. Work submitted through email, Canvas Inbox, or an incorrect assignment folder may receive no credit. Unless otherwise directed, analytical assignments requiring graphs and calculations must include the RStudio project, code, outputs, and supporting written interpretation.

Assessment and Grading

Assessment	Points	Evidence of Learning
Problem Set 1	70	Demand/supply analysis, equilibrium calculations, and R visualization.
Problem Set 2	80	Market-failure analysis and alternative policy options.
Midterm Examination	80	Microeconomic concepts and applied analysis.
Final Examination	80	Cumulative economic and data-science analysis.
RStudio / Cloud Learning Portfolio	30	Setup completion, starter scripts, reproducible outputs, and troubleshooting activities.
Weekly Discussions	60	Four evidence-based discussions and constructive peer engagement.
Total	400	

Final grading scale: A = 360-400; B = 320-359; C = 280-319; F = below 280.

Problem Sets and Final Project Deliverable

Problem sets and selected exam tasks apply economics to public-sector problems. A recurring workflow will be used:

- Start with demand and supply equations to compute equilibrium points.
- Use approved AI support, where authorized, to explain or refine R code; validate and document the result.
- Create graphical visualizations in RStudio using Jetstream2 or CloudBank.
- Interpret the results, identify implications for public policy, and propose alternatives for a market failure.
- Submit the required Canvas deliverables, including code, outputs, and written analysis, and use the grading rubric to self-check before submission.

The final project/deliverable is an integrated policy-data analysis that uses an approved public dataset, reproducible R code, visualizations, and a policy recommendation. Detailed prompts and rubric will be posted in Canvas.

Discussion and Participation

Students will complete four weekly discussions. Each discussion requires an evidence-based response and a constructive evaluation of at least one colleague's post. Use APA-style in-text citations and references. Regular, punctual attendance and active participation are expected. Students should arrive prepared to discuss assigned readings, methods, and policy implications.

Late Work, Make-Up Work, and Withdrawal

Late submissions for Problem Sets, weekly discussions, and other eligible work are subject to a five-point deduction for each day after the deadline until the work is graded, unless the instructor approves an exception.

No late final examination is permitted. Make-up work is available only for reasons accommodated by the University. Students are responsible for observing University withdrawal deadlines; failure to withdraw may result in a final grade of F.

Course Schedule

The instructor may modify the schedule, readings, assignments, or technical workflow as needed. Specific dates and deadlines will be posted in Canvas.

Class	Module / Topic	Reading, Activity, or Due Work
1	Module 1: Introduction to Economics and Data Science	Callan/Thomas Ch. 1; Robbins Ch. 1; Data Science Ch. 1; Cloud access orientation.
2	Module 1: Introduction to Economics and Data Science	Weekly Discussion 1; responsible-AI and data-source orientation.
3	Module 2: Nature of Goods	Ostrom Ch. 7; R/RStudio onboarding lab.
4	Module 2: Nature of Goods	Ostrom Ch. 7; Weekly Discussion 2; Jetstream2/CloudBank troubleshooting lab.
5	Module 3: R Basics and Data Visualization	Introduction to Data Science Chs. 2-11; starter R script and visualization lab.
6	Module 3: R Basics and Data Visualization	Midterm examination due; reproducible-project check.
7	Module 3: R Basics and Data Visualization	Public dataset exploration; data-storytelling practice.
8	Module 4: Market Provision of Goods	Callan/Thomas Ch. 2; tax/subsidy impact and areas of plane shapes.
9	Module 4: Market Provision of Goods	Problem Set 1 due; R graph submission.
10	Module 5: Solutions to Market Failure	Callan/Thomas Chs. 4-5; policy-options workshop.
11	Module 5: Solutions to Market Failure	Problem Set 2 due; cloud-based analysis check.
12	Module 6: Wealth Distribution	Robbins Ch. 16; Weekly Discussion 3.
13	Module 7: International Trade	Robbins Ch. 17; final-project data and methods checkpoint.
14	Module 7: International Trade	Robbins Ch. 17; Weekly Discussion 4.
15	Module 8: Ethical Considerations in Data Science	Algorithmic bias, privacy, provenance, and responsible AI.
16	Module 8: Final Integration	Final project/deliverable and final examination.

Academic Integrity, Accessibility, and Student Support

Academic dishonesty, including plagiarism, falsification, unauthorized collaboration, or misrepresentation of AI assistance, is inconsistent with Clark Atlanta University policy and may result in penalties up to a failing grade and further University action. Use APA-style citation and acknowledge all sources, data, software, and authorized AI assistance as required.

Clark Atlanta University is committed to providing equal access for students with documented disabilities. Students seeking reasonable accommodations should contact the Office of Counseling and Disability Services at 404-880-8044, Trevor Arnett Hall, 3rd floor, and provide the instructor with the applicable Letter of Accommodation.

The course is grounded in the University's core values: student centeredness, pursuit of excellence, innovation and discovery, integrity, social responsibility, and respect. Students are expected to contribute to a learning environment characterized by honesty, mutual respect, and professional engagement.